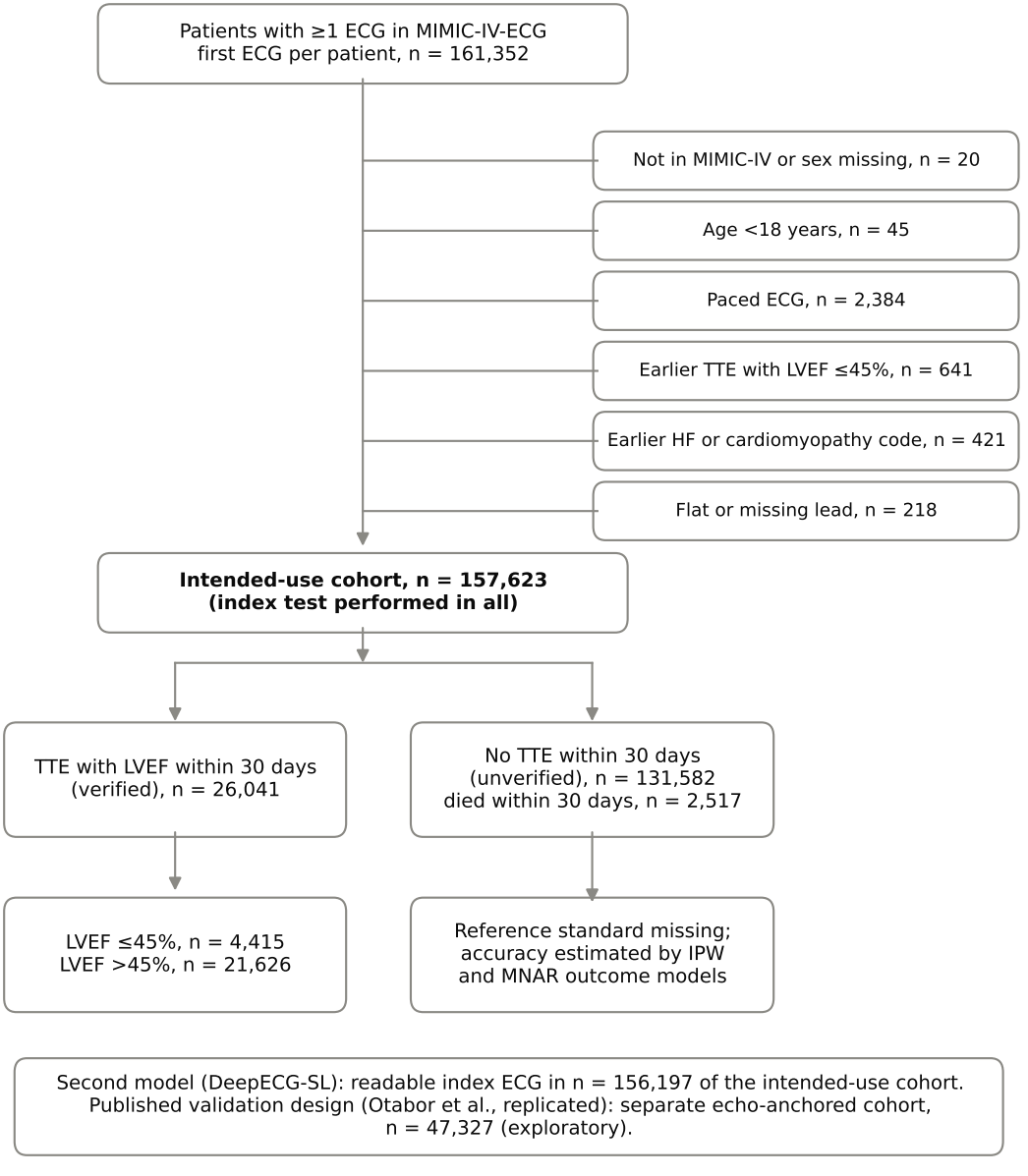
Patients with ≥1 ECG in MIMIC-IV-ECG
first ECG per patient, n = 161,352
Not in MIMIC-IV or sex missing, n = 20
Age <18 years, n = 45
Paced ECG, n = 2,384
Earlier TTE with LVEF ≤45%, n = 641
Earlier HF or cardiomyopathy code, n = 421
Flat or missing lead, n = 218
Intended-use cohort, n = 157,623
(index test performed in all)
TTE with LVEF within 30 days
(verified), n = 26,041
No TTE within 30 days
(unverified), n = 131,582
died within 30 days, n = 2,517
LVEF ≤45%, n = 4,415
LVEF >45%, n = 21,626
Reference standard missing;
accuracy estimated by IPW
and MNAR outcome models
Second model (DeepECG-SL): readable index ECG in n = 156,197 of the intended-use cohort.
Published validation design (Otabor et al., replicated): separate echo-anchored cohort,
n = 47,327 (exploratory).